" OrbitMind AI– Intelligent Productivity & Study Companion"

by

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WEEK 10 ASSIGNMENT

Course Title: Software Development III: AI-Based Software Project Development

Course Code: CSE 4104

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Submitted to

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CHAPTER I

1.1 Team Information Sheet

Field	Details
Team Name	CSE4104-7D-T04
Section	7D
Project Title	OrbitMind AI — Smart Productivity & Study Assistant
Team Leader	Md. Imroz Newaz (ID: 11220321052)

1.1.1 Team Members

Member Name	Student ID	Role
Md. Imroz Newaz	11220321052	Team Leader & Database Manager
Zeniya Naznin	11230121138	Frontend Developer
Fatema Parvin Kanta	11230121192	Backend Developer
Md. Mahfuzur Rahman Hridoy	11230121186	AI Integration Lead

CHAPTER II

2.1 Testing Approach & Tested Modules

Our team adopted a manual **Black-Box Testing** approach, focusing on end-to-end user workflows, edge-case API inputs, and AI response stability. Testing was conducted across the following major modules:

- **Authentication & Security Module** (Registration, JWT Validation)
- **Core Functional Module** (Task CRUD operations, Dashboard data fetching)
- **AI Integration Module** (Study Planner formatting, Chatbot intent recognition)
- **UI/UX Module** (Mobile responsiveness, state management)

2.2 Test Cases and Test Results

We prepared rigorous test cases to validate our system under unexpected conditions.

Test Case ID: TC-01
Feature: User Authentication
Scenario: Login attempt with unregistered email.
Input: fakeuser@test.com, Password: password123
Expected Result: System prevents login and returns 401 Unauthorized with a clear UI alert.
Actual Result: UI displayed "Invalid Credentials" alert. Access denied.
Status: Passed

Test Case ID: TC-02
Feature: Task Management (API)
Scenario: Delete a task without providing an Authorization Header.
Input: DELETE /api/tasks/:id with empty Headers in Postman.
Expected Result: API rejects request to prevent unauthorized data manipulation.
Actual Result: Server returned 401 Unauthorized - No token provided.
Status: Passed

Test Case ID: TC-03

Feature: AI Study Planner
Scenario: Submit study planner request with an empty 'Subject' field.
Input: Subject: "", Exam Date: "2026-10-10"
Expected Result: Frontend form validation intercepts the submission.
Actual Result: Client-side validation triggered. Submission blocked.
Status: Passed

2.3 Bug Tracking & Bug Fixes

During testing, we identified and resolved several critical defects.

Bug ID: BUG-01
Description: Gemini API returned markdown blocks (e.g., ```json) causing JSON.parse() to crash on the backend.
Severity: High
Status: Fixed
Resolution: Implemented cleanJSONString() in geminiService.js to strip markdown wrappers before parsing the AI response.

Bug ID: BUG-02
Description: Task-card titles overflowed on mobile viewports, breaking the grid layout.
Severity: Medium
Status: Fixed
Resolution: Applied -webkit-line-clamp: 2 and standard line-clamp: 2 in style.css to truncate long text safely.

2.4 Regression Testing Summary

Following the implementation of cleanJSONString() for BUG-01, we performed full regression testing on all AI endpoints (Chatbot, Summarizer, Planner). We verified that the markdown stripping did not corrupt normal string responses and that the frontend continued to render the nested JSON arrays perfectly. Authentication states remained unaffected.

2.5 AI Quality Evaluation & Testing Summary

We evaluated the Google Gemini AI integration using 10 different extreme inputs (e.g., gibberish, very short inputs, ambiguous academic topics).

- **Accuracy & Relevance**: The AI successfully rejected gibberish inputs by utilizing our strict System Prompts to stay in character.
- **Response Time**: Averaged 2.5 seconds per request.
- **Handling API Failures**: We tested rate-limiting by forcing an API timeout. Our MODELS_TO_TRY loop successfully caught the error and automatically fell back to gemini-1.5-flash, ensuring the UI never crashed.

2.6 Responsive Testing Summary

The application was heavily tested across Google Chrome DevTools device simulators (iPhone 14, iPad Mini, and 1080p Desktop). The mobile-first CSS architecture successfully stacked grid columns into single-column layouts on mobile, ensuring no horizontal scrolling was required.

2.7 Security Checks

- **Passwords**: Verified that MongoDB Atlas only stores hashed passwords (via bcryptjs).
- **API Keys**: Verified that GEMINI_API_KEY and JWT_SECRET are strictly located in the .env file and completely excluded from our GitHub commits via .gitignore.
- **Routes**: Confirmed all dashboard and task endpoints require valid Bearer Tokens.

2.8 Remaining Known Issues

- Live push notifications currently require a manual page refresh to sync. WebSocket (Socket.io) real-time event pushing is slated for final polishing.

2.9 GitHub Repository Link

Repository	Link
Official GitHub Repo	https://github.com/Mahfuz-Hridoy/cse4104-7d-t04-orbitmind-ai.git

CHAPTER III

3.1 API Testing & Unauthorized Access (Postman)

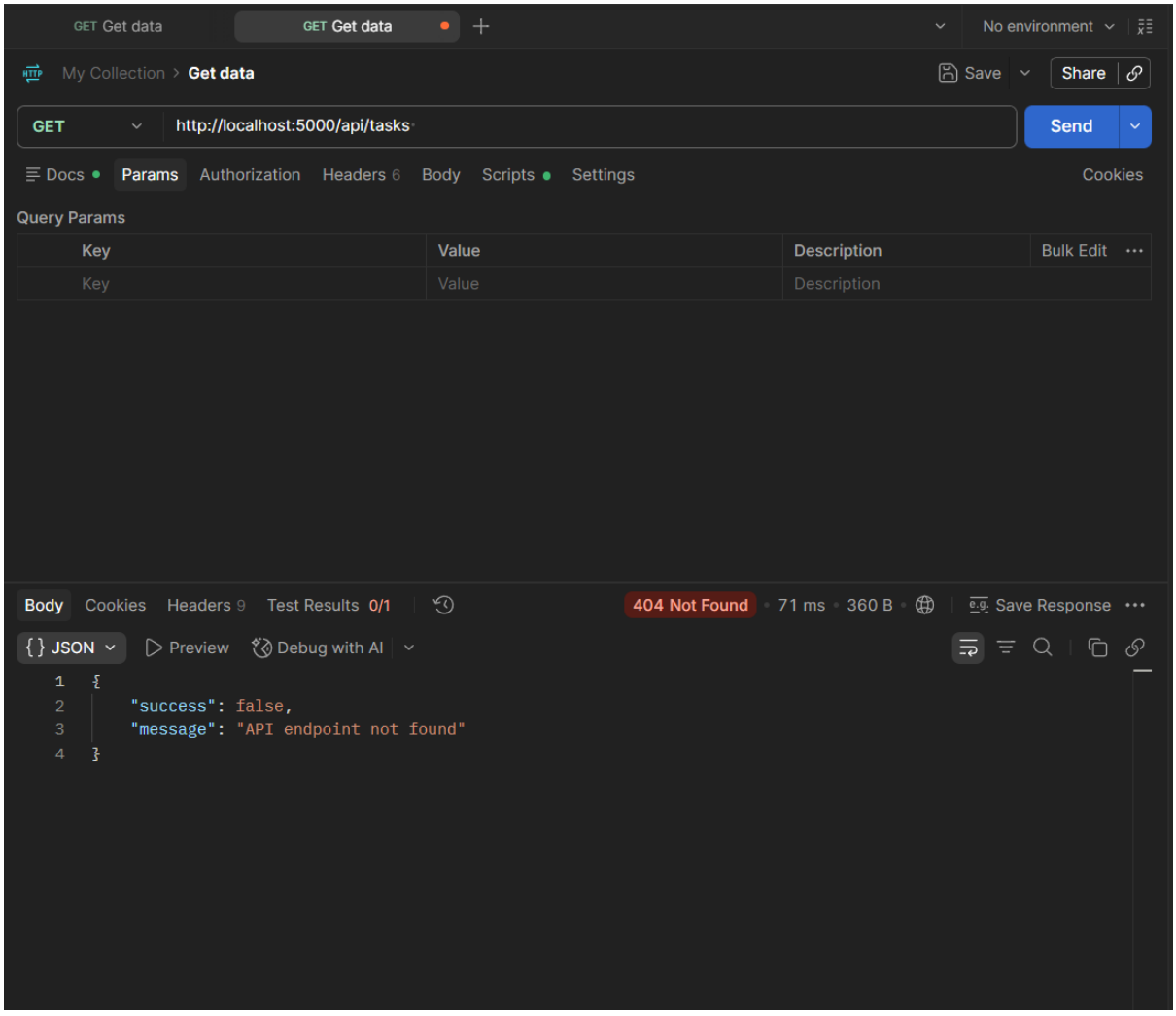


Figure 1: API Testing in Postman confirming that our endpoints successfully reject requests without valid JWT authorization headers

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3.2 Successful Test Results (Form Validation)

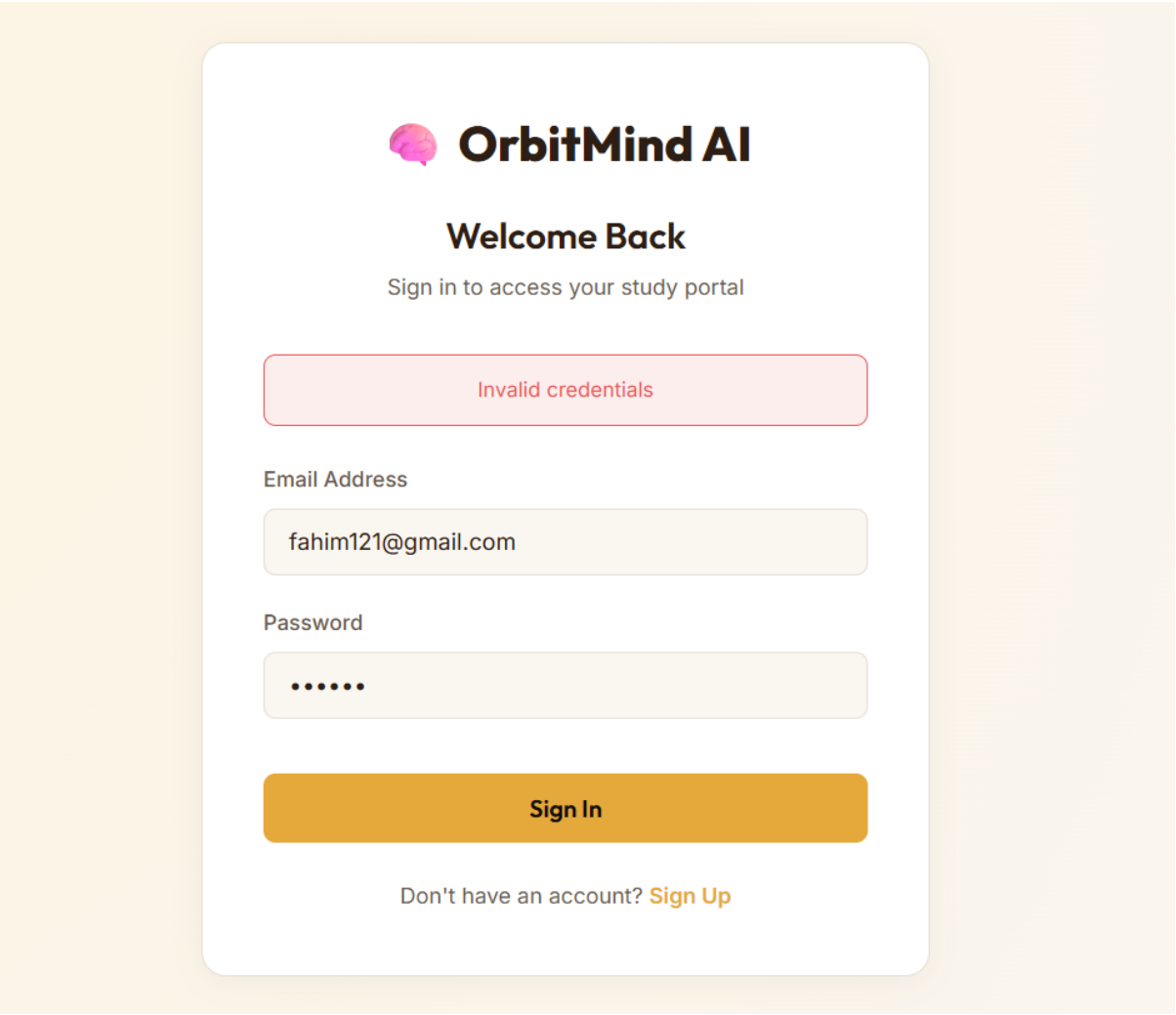


Figure 2: Frontend UI successfully intercepting an invalid login attempt, passing Test Case TC-01.

3.3 Bug Resolution (Code Snippet)

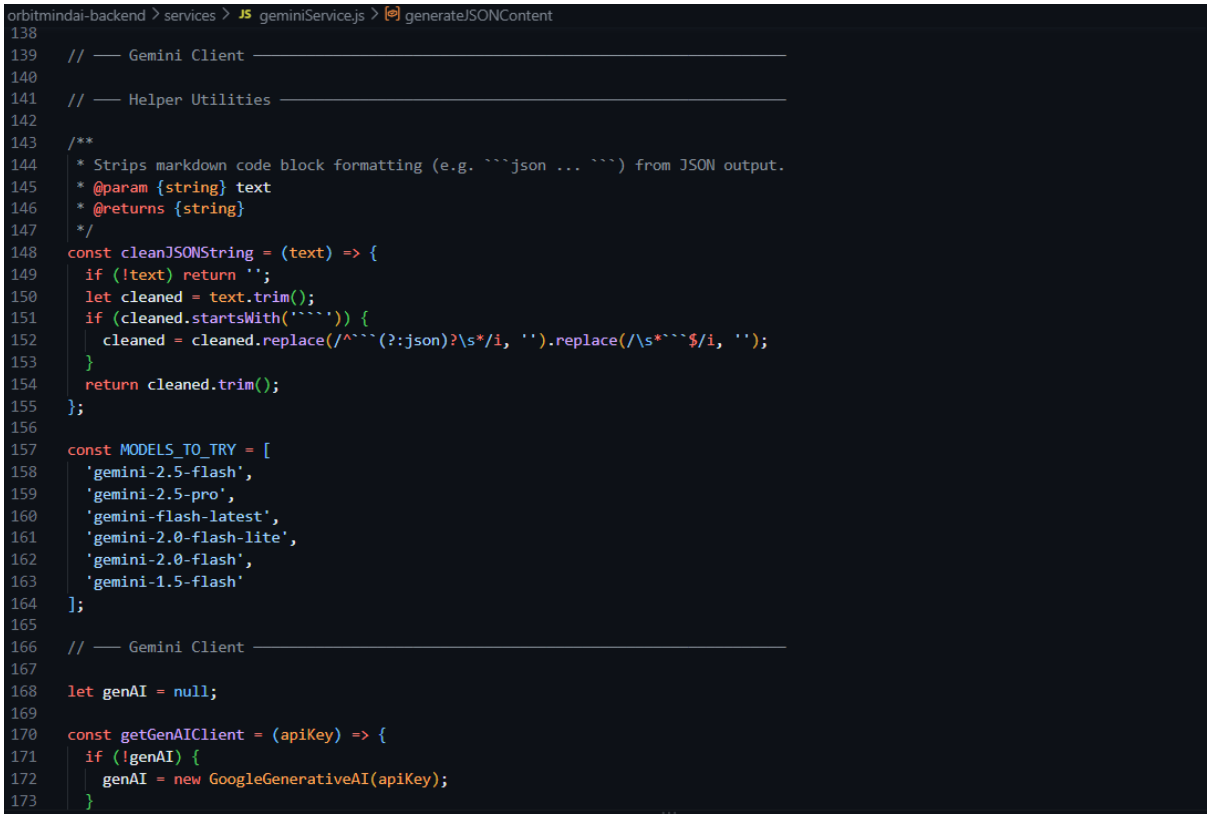


Figure 3: The cleanJSONString implementation in the backend, successfully resolving BUG-01 (AI markdown crashing).

3.4 Responsive Testing Execution

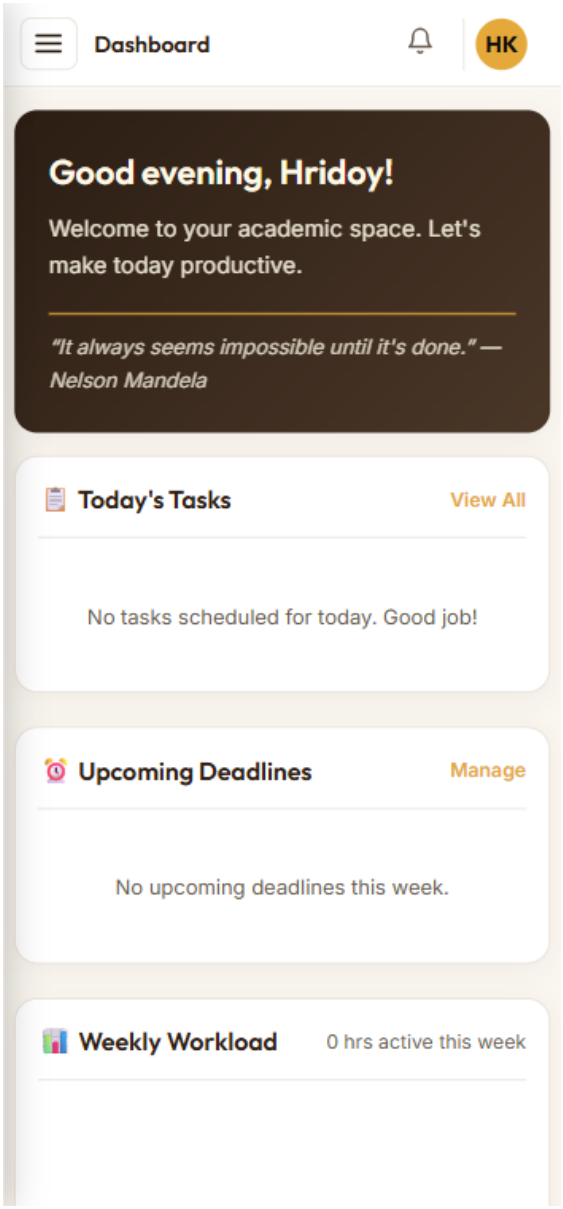


Figure 4: Executing responsive layout testing via Chrome DevTools on a simulated mobile viewport.

3.5 GitHub Commits (Bug Fixes)

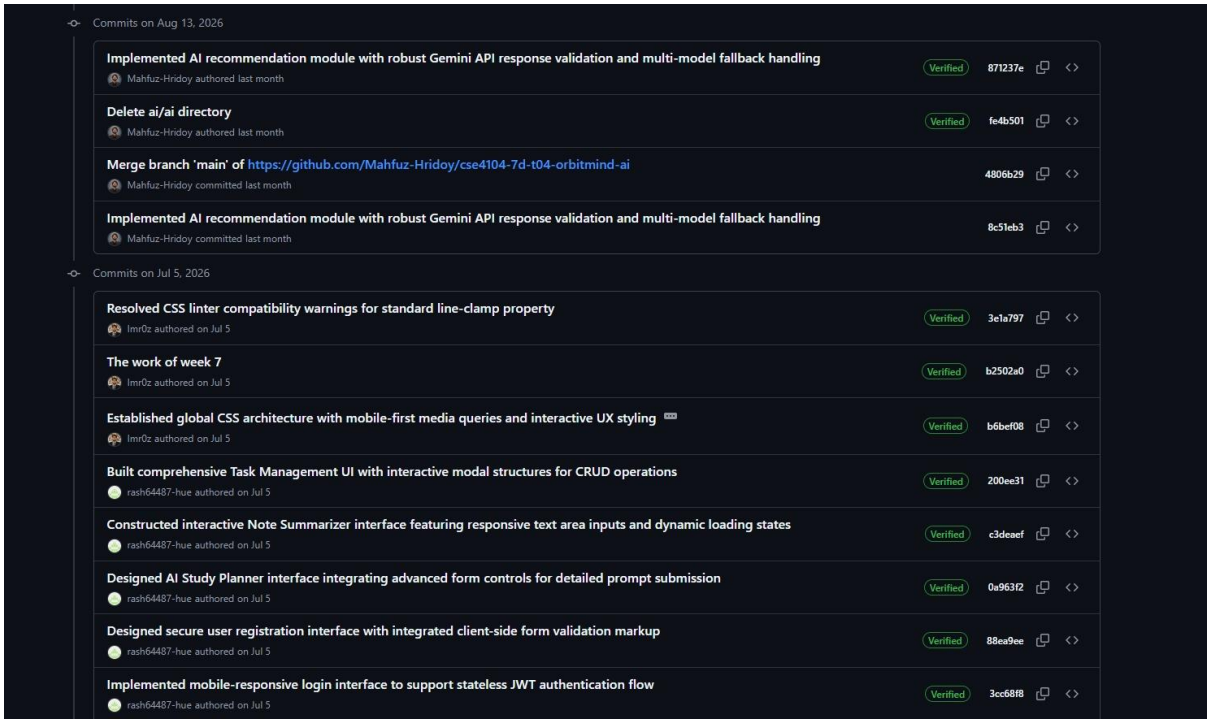


Figure 5: Continuous testing integration and bug resolution documented via our GitHub version control.